

**Exam 5 Review**

## (1) Line Integrals

- Consider the function  $f(x, y) = x - y$  and the curve  $C$  which connects the points  $(1, 1)$  to  $(2, 3)$  linearly.

a) Fully setup an integral to represent  $\int_C f(x, y) ds$ .

b) Calculate  $\int_C f(x, y) dx$ .

c) Calculate  $\int_C f(x, y) dy$ .

- Consider the vector field  $\mathbf{F}(x, y) = \left\langle xy, \frac{1}{2}x^2 \right\rangle$  and the curve C which traces outlines the region in the first quadrant determined by  $y = x^2$  &  $y = x$  in a counterclockwise fashion.
- a) Determine  $\int_C \mathbf{F} \cdot d\mathbf{r}$  using techniques from section 5.2.

- b) Determine  $\int_C \mathbf{F} \cdot d\mathbf{r}$  using the Fundamental Theorem of Line Integrals.

- c) Determine  $\int_C \mathbf{F} \cdot d\mathbf{r}$  using Green's Theorem.

## (2) Surface Integrals

- Consider the function  $f(x, y, z) = xyz$  and the surface  $\sigma$  given by  $z = x^2 + y^2$  &  $z = 9$  which encloses a solid region  $R$  and is oriented outward.
- a) Set up, but do not evaluate, an integral to compute  $\iint_{\sigma} f(x, y, z) dS$  using a parameterization of  $\sigma$ .

- b) Set up, but do not evaluate, an integral to compute  $\iint_{\sigma} f(x, y, z) dS$  without using a parameterization of  $\sigma$ .

- c) Set up, but do not evaluate, an integral to compute  $\iint_{\sigma} \mathbf{F} \cdot d\mathbf{S}$  by calculating the flux through each piece of the surface where  $\mathbf{F} = \langle 1, x, z \rangle$ .

- d) Set up, but do not evaluate, an integral to compute  $\iint_{\sigma} \mathbf{F} \cdot d\mathbf{S}$  using the Divergence Theorem.

### (3) Stokes' Theorem

- a) Compute  $\iint_{\sigma_1} \text{curl}(\mathbf{F}) \cdot d\mathbf{S}$  where  $\sigma_1$  is the portion of the surface  $\sigma$  given by  $z = x^2 + y^2$  directly. Assume the surface is oriented with upward facing normal and  $\mathbf{F} = \langle 1, x, z \rangle$ .

- b) Compute  $\iint_{\sigma_1} \text{curl}(\mathbf{F}) \cdot d\mathbf{S}$  where  $\sigma_1$  is the portion of the surface  $\sigma$  given by  $z = x^2 + y^2$  using Stokes's Theorem. Assume the surface is oriented with upward facing normal and  $\mathbf{F} = \langle 1, x, z \rangle$ .

- c) Set up, but do not evaluate, and integral to compute  $\int_C \mathbf{F} \cdot d\mathbf{r}$  where  $\mathbf{F}(x, y) = \left\langle y, \frac{1}{2}x^2, 1 \right\rangle$  and  $C$  is the boundary curve of the surface  $\sigma$  given by  $z = 1 - x^2$  (with upward orientation) over the region  $D = \{(x, y) | x^2 \leq y \leq x \text{ \& } 0 \leq x \leq 1\}$  using Stokes' Theorem.